Type MS · Getting from A to B: subtracting position vectors

Answer 1 of 8 · Question 1 part a · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

1(a)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}, = -3\mathbf{i} + 4\mathbf{j}$	M1,A1 (2)
(b)	$\left\|\overrightarrow{AB}\right\| = 5$	
	Unit vector $= \frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ or $-\frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ oe Accept column vectors	M1A1 (2)
		[4]

(a)	
M1	For $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ seen, or $\overrightarrow{AB} = (\mathbf{i} + 7\mathbf{j}) - (4\mathbf{i} + 3\mathbf{j})$ or equivalent in column form
A1	Correct simplified answer as shown or equivalent but NOT a column vector
(b)	
M1	Correct modulus of their $\overrightarrow{AB}$ **and** divide ± their $\overrightarrow{AB}$ by it
A1	Correct unit vector, as shown or equivalent inc column vector $\pm\frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ scores M1A0 $\frac{1}{5} - 3\mathbf{i} + 4\mathbf{j}$ scores M1A0
NB:	If $\overrightarrow{BA}$ is found in (a) both (b) marks are still available

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Question Number	Scheme	Marks
	$\overrightarrow{d4}$	

Type MS · Getting from A to B: subtracting position vectors

Answer 2 of 8 · Question 7 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

number		
7 (a)	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$ $\overrightarrow{BC} = -(15\mathbf{i} - 6\mathbf{j}) + 8\mathbf{i} + \mathbf{j} = -7\mathbf{i} + 7\mathbf{j}$	M1 A1 [2]
(b)	$\left\lvert\overrightarrow{BC}\right\rvert = \sqrt{98} = \left(7\sqrt{2}\right)$ Unit vector is $\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe	B1 B1 [2]
(c)	$\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{MC}$ are parallel oe (and have same point of origin (M)) hence they are collinear. **ALT 1** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $\Rightarrow \overrightarrow{NC} = -(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{NC}$ are parallel oe (and share the same point (N)) hence they are collinear. **ALT 2** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MC}$ and $\overrightarrow{NC}$ are parallel oe(and share the same point (C)) hence they are collinear.	B1 M1 M1 A1 [4] {B1} {M1} {M1} {A1} [4] {B1} {M1} {M1} {A1} [4]
		Total 8 marks

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(a)	
M1	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$
A1	$\overrightarrow{BC} = -7\mathbf{i} + 7\mathbf{j}$
(b)	
B1	$\sqrt{98}$ oe
B1	$\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe
(c)	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ (may be implied by $\overrightarrow{MN}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{MC} = 4\overrightarrow{MN}$
ALT 1	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{MN}$ or $\overrightarrow{NC}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $-(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = 3\overrightarrow{MN}$
ALT 2	
B1	$\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{NC}$)
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = \frac{3}{4}\overrightarrow{MC}$
	For part c: Send any geometrical solutions to review

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Question number	Scheme	Marks

Type MS · Getting from A to B: subtracting position vectors

Answer 3 of 8 · Question 1 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

1. Factorisation:

$\left(x^2+bx+c\right)=\left(x+p\right)\left(x+q\right)$, where $|pq|=|c|$ leading to $x=....$

$\left(ax^2+bx+c\right)=\left(mx+p\right)\left(nx+q\right)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

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Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

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Answer 3 of 8 · Question 1 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

International GCSE Further Pure Mathematics – Paper 2R mark scheme

Question number	Scheme	Marks
1(a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$ or $6\mathbf{i} + 8\mathbf{j} = \overrightarrow{OB} - (3\mathbf{i} - 2\mathbf{j})$ oe $9\mathbf{i} + 6\mathbf{j}$	M1 A1 [2]
(b)	$\sqrt{6^2 + 8^2}$ *or* 10 (from Pythagorean triple)	B1 [1]
(c)	$(\pm)\frac{1}{"10"}(6\mathbf{i} + 8\mathbf{j})$	M1 A1 [2]
		Total 5 marks

Type MS · Getting from A to B: subtracting position vectors

Answer 4 of 8 · Question 10 part a.i · 0 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a.i in this mark scheme extract

10(a)	(i) $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	B1
	(ii) $\overrightarrow{MY} = \overrightarrow{MA} + \frac{3}{4}\left(\overrightarrow{AB}\right) = \mathbf{a} + \frac{3}{4}(-2\mathbf{a} + 4\mathbf{b}) = -\frac{\mathbf{a}}{2} + 3\mathbf{b}$	M1A1 [4]
(b)	$\overrightarrow{OX} = \mu\overrightarrow{OB} = \mu 4\mathbf{b}$	M1
	$\overrightarrow{OX} = \overrightarrow{OM} + \overrightarrow{MX} = \overrightarrow{OM} + \lambda\overrightarrow{MY} = \mathbf{a} + \lambda\left(-\frac{\mathbf{a}}{2} + 3\mathbf{b}\right) = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\lambda\mathbf{b}$	M1
	$\Rightarrow \mu 4\mathbf{b} = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\mathbf{b}$	dM1
	$\Rightarrow 1 - \frac{\lambda}{2} = 0 \Rightarrow \lambda = 2$	ddM1
	$\Rightarrow 4\mu = 3\lambda \Rightarrow \mu = \frac{6}{4} = \frac{3}{2}$	
	$OB : OX = 2:3$ oe	A1 [5]
	ALT – working with alternative vector within triangle *OMX*	
	$\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OB} + \mu\overrightarrow{OB} = -\boldsymbol{a} + 4\boldsymbol{b} + \mu 4\boldsymbol{b}$	[M1
	$\overrightarrow{MX} = \lambda\left(-\frac{\boldsymbol{a}}{2} + 3\boldsymbol{b}\right)$	M1
	$\Rightarrow -\boldsymbol{a} = -\frac{\lambda\boldsymbol{a}}{2} \Rightarrow \lambda = 2$	dM1
	$\Rightarrow 4\boldsymbol{b} + \mu 4\boldsymbol{b} = 3\lambda\mathbf{b} \Rightarrow \mu = \frac{1}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1]
(c)	$\frac{\Delta YBX}{\Delta ABX} = \frac{1}{4}$	M1
	$\frac{\Delta ABX}{\Delta OAX} = \frac{1}{3}$	
	$\Rightarrow \frac{\Delta YBX}{\Delta OAX} = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} \Rightarrow \Delta YBX : \Delta OAX = 1:12$	M1A1 [3]
	ALT – working with relative areas of triangles	
	Area $\Delta YBX = a$ Area $\Delta ABX = 4a$ Area $\Delta OMX = 6a$ Area $\Delta AYX = 3a$ Area $\Delta OAY = 6a$ Area $\Delta OYB = 2a$	[M1
	Area $\Delta OAX = 12a$	M1
	$\Delta YBX : \Delta OAX = 1:12$	A1]
		Total 11 marks

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Part	Mark	Notes
(a)	B1	For the correct simplified vector $\overrightarrow{AB}$
	M1	For the correct vector statement for $\overrightarrow{MY}$
	A1	For the correct simplified vector for $\overrightarrow{MY}$
(b)	M1	For the statement $\overrightarrow{OX} = \mu 4\mathbf{b}$ Note: this is a B mark on epen.
	M1	For the correct vector for $\overrightarrow{OX}$ (ft their MY)
	dM1	For equating both vectors for $\overrightarrow{OX}$ and for comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5
	ALT – working with alternative vector within triangle OMX	
	M1	For a vector statement which includes $\mu 4\boldsymbol{b}$ (for OX or BX) (ft their MY) Note: this is a B mark on epen.
	M1	For a correct second vector equation for the same vector (ft their MY)
	dM1	For equating both vectors and comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5 Condone $OX:OB = 3:2$ if clearly stated, but not just 3:2
(c)	M1	For either the relationship between the of areas of triangles BYX and ABX **or** the relationship between the areas of triangles ABX and OAX
	M1	For finding the relationship between the of areas of triangles BYX and OAX
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).
	ALT – working with relative areas	
	M1	For assigning a value to one triangle area and writing a second area in terms of this. Note: This could also follow from working with area of a triangle $= \frac{1}{2}ab\sin C$

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Answer 4 of 8 · Question 10 part a.i · 0 marks

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		e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	M1	For finding the relationship between the of areas of triangles BYX and OAX Note: This could also follow from working with area of a triangle $= \frac{1}{2} ab \sin C$ e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).

Type MS · Getting from A to B: subtracting position vectors

Answer 4 of 8 · Question 10 part a.i · 0 marks

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Type MS · Getting from A to B: subtracting position vectors

Answer 5 of 8 · Question 8 part a.i · 0 marks

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8 (a) (i)	$\overrightarrow{(AB)} = \overrightarrow{AO} + \overrightarrow{OB}\ (= -4\mathbf{a} - 5\mathbf{b} + 8\mathbf{a} - \mathbf{b}) = 4\mathbf{a} - 6\mathbf{b}$	M1A1 [2]
(ii)	$\left\{ \left\| \overrightarrow{AB} \right\| = \sqrt{52} \right\} = 2\sqrt{13}$	B1 [1]
(b)	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2}\ (= \sqrt{41})$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2}\ (= \sqrt{65})$	M1
	$\left(\sqrt{52}\right)^2 = \left(\sqrt{41}\right)^2 + \left(\sqrt{65}\right)^2 - 2 \times \sqrt{41} \times \sqrt{65} \cos(AOB)$	M1
	$\left\{ \text{angle } AOB = \cos^{-1}\left(\frac{41 + 65 - 52}{2 \times \sqrt{41} \times \sqrt{65}} \right) = 58.465... \right\}$	
	Area triangle $AOB = \frac{1}{2} \times "\sqrt{41}" \times "\sqrt{65}" \times \sin "58.465..."$	M1
	$= 22$ awrt	A1 [4]
ALT 1	Area $= \frac{1}{2} \begin{vmatrix} 0 & 4 & 8 & 0 \\ 0 & 5 & -1 & 0 \end{vmatrix}$ oe	M1
	$= \frac{1}{2} \left\| (0 \times 5 + 4 \times (-1) + 8 \times 0) - (0 \times 4 + 5 \times 8 + (-1) \times 0) \right\|$ oe	M1
	$\left\{ = \frac{1}{2} \left\| -44 \right\| \right\} = 22$	M1A1 [4]
ALT 2	Use Heron's formula	
	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2}\ (= \sqrt{41})$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2}\ (= \sqrt{65})$	M1
	$p = \frac{"\sqrt{41}" + "\sqrt{65}" + "\sqrt{52}"}{2}$	M1
	Area $= \sqrt{p(p - "\sqrt{41}") \times (p - "\sqrt{65}") \times (p - "\sqrt{52}")}$	M1
	$= 22$ exact	A1 [4]
(c)	$\overrightarrow{OC} = \lambda \overrightarrow{OD} \left\{ = \lambda(15\mathbf{a} + 10\mathbf{b}) \right\}$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu \overrightarrow{AB} \left\{ = 4\mathbf{a} + 5\mathbf{b} + \mu(4\mathbf{a} - 6\mathbf{b}) = (4 + 4\mu)\mathbf{a} + (5 - 6\mu)\mathbf{b} \right\}$	M1
	$4 + 4\mu = 15\lambda$ *and* $5 - 6\mu = 10\lambda \Rightarrow \mu = ...$ *or* $\lambda = ...$	ddM1
	$\mu = \frac{7}{26}$ *or* $\lambda = \frac{22}{65}$	A1
	$\overrightarrow{OC} = \frac{66}{13}\mathbf{a} + \frac{44}{13}\mathbf{b}$ or $5\frac{1}{13}\mathbf{a} + 3\frac{5}{13}\mathbf{b}$ cao	A1 [5]
		Total 12 marks

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Part	Mark	Additional Guidance
(a)(i)	M1	For a correct vector path to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ or for $\pm(-4\mathbf{a}-5\mathbf{b}+8\mathbf{a}-\mathbf{b})$
	A1	For the correct vector $4\mathbf{a}-6\mathbf{b}$
(ii)	B1	For the correct exact simplest magnitude $2\sqrt{13}$
(b)	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For correct substitution into the cosine rule. e.g. $(\sqrt{65})^2=(\sqrt{41})^2+(\sqrt{52})^2-2\times\sqrt{41}\times\sqrt{52}\cos(OAB)$ or $(\sqrt{41})^2=(\sqrt{52})^2+(\sqrt{65})^2-2\times\sqrt{52}\times\sqrt{65}\cos(ABO)$
	M1	For correct substitution into the area of a triangle formula, use their values. e.g. $\frac{1}{2}\times\text{"}\sqrt{41}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}72.349...\text{"}$ or $\frac{1}{2}\times\text{"}\sqrt{65}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}49.184...\text{"}$
	A1	awrt 22
ALT1	M1	For a correct statement for the area, must have $\frac{1}{2}$ (Start with any point, go clockwise or anticlockwise, but must repeat the first point at the end.)
	M1	For a correct evaluation of their determinant, ignore $\frac{1}{2}$
	M1	$\frac{1}{2}$ multiplies their determinant value, correct answer implies this mark.
	A1	Correct **exact** value 22, **not awrt**
ALT2	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For finding half of the perimeter
	M1	Substitute the perimeter and 3 lengths to a correct Heron's formula
	A1	Correct **exact** value 22, **not awrt**
(c)	M1	For $\overrightarrow{OC}=\lambda\overrightarrow{OD}$ (can be implied in their later work)
	M1	For correct vector statement $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}$ oe Can be implied for later substitution of their vectors **Please note**: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{BA}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{AB}$ is also correct, provided they have found a negative value of μ
	ddM1	Equating components and reaching a value for either of their parameters. Dependent on both previous method marks
	A1	Either correct value of their parameters. They do not need to work out both values. Note: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}\Rightarrow\mu=\frac{7}{26}$, $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}\Rightarrow\mu=\frac{19}{26}$
	A1	Correct vector in simplified expression

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Question number	Scheme	Marks

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Answer 6 of 8 · Question 4 part b · 2 marks

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Question	Scheme	Marks
4(a)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} \Rightarrow \overrightarrow{AB} = (5\mathbf{i} + 9p\mathbf{j}) - (p\mathbf{i} + 2p\mathbf{j})$	M1A1
	$\mathbf{i}(5-p) + \mathbf{j}(7p) = Q(\mathbf{i} - 2\mathbf{j}) \Rightarrow 5 - p = Q$ and $7p = -2Q$	M1M1
	$7p = -2(5-p) \Rightarrow p = -2$	M1A1 [6]
(b)	$7('-2') = -2Q \Rightarrow Q = 7$, $\overrightarrow{AB} = 7(\mathbf{i} - 2\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	M1A1ft [2]
	OR	
	$\overrightarrow{AB} = (5\mathbf{i} + 9(-2)\mathbf{j}) - ((-2)\mathbf{i} + 2(-2)\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	[M1A1ft]
(c)	$\overrightarrow{OA} = -2\mathbf{i} - 4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2 + (-4)^2} = \sqrt{20}$	M1A1ft
	Unit vector is $\frac{1}{\sqrt{20}}(-2\mathbf{i} - 4\mathbf{j}) = \frac{\sqrt{5}}{5}(-\mathbf{i} - 2\mathbf{j})$	M1A1 [4]
		Total 12 marks

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Answer 6 of 8 · Question 4 part b · 2 marks

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Question	Notes	Marks
4(a)	For the basic vector statement $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$	M1
	For the correct vector (simplified or unsimplified) $\overrightarrow{AB} = (5\mathbf{i}+9p\mathbf{j}) - (p\mathbf{i}+2p\mathbf{j}) = [\mathbf{i}(5-p)+\mathbf{j}(7p)]$	A1
	For setting their $\overrightarrow{AB} = Q(\mathbf{i}-2\mathbf{j})$ where $Q \neq 1,\ Q \neq 0$ $\mathbf{i}(5-p)+\mathbf{j}(7p) = Q(\mathbf{i}-2\mathbf{j})$	M1
	For equating components of **i** and **j** **i** $\quad 5-p = Q$ **j** $\quad 7p = -2Q$	M1
	Solving the simultaneous equations by any method to find the value of p $7p = -2(5-p) \Rightarrow p = ...$	M1
	For the value of $p = -2$	A1 [6]
(b)	For finding the value of k and using it to find the vector $\overrightarrow{AB}$ $7('-2') = -2Q \Rightarrow Q = 7$	M1
	For the correct vector $\overrightarrow{AB} = 7(\mathbf{i}-2\mathbf{j}) = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
	ALT	
	For substituting their value of p to find the vector $\overrightarrow{AB}$ $\overrightarrow{AB} = \mathbf{i}(5-[-2])+\mathbf{j}(7[-2]) = ...$	M1
	$\overrightarrow{AB} = \mathbf{i}(5-p)+\mathbf{j}(7p) \Rightarrow \overrightarrow{AB} = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
(c)	$\overrightarrow{OA} = -2\mathbf{i}-4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2+(-4)^2} = ...$	M1
	$\left\lvert\overrightarrow{OA}\right\rvert = \sqrt{20}$	A1ft
	Unit vector in the direction of $\overrightarrow{OA}$ is $\frac{1}{\sqrt{20}}(-2\mathbf{i}-4\mathbf{j})$	M1
	Unit vector in the required form $\frac{\sqrt{5}}{5}(-\mathbf{i}-2\mathbf{j})$ Allow $\frac{\sqrt{5}}{5}(\mathbf{i}+2\mathbf{j})$ **provided** no processing errors seen.	A1 [4]
		Total 12 marks

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Answer 6 of 8 · Question 4 part b · 2 marks

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Question	Scheme	Marks

Type MS · Getting from A to B: subtracting position vectors

Answer 7 of 8 · Question 9 part a · 2 marks

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9(a)	$\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} \Rightarrow \overrightarrow{OP} = 2\mathbf{a} + \frac{1}{2}(4\mathbf{b} - 2\mathbf{a}) = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(c)	$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}(\overrightarrow{OP})$ $\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a} + 2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$	M1 B1 A1 [3]
(d)	$\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$ $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$ $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \left(\frac{8}{5}\right)$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \frac{12}{20}, \left(\frac{3}{5}\right)$ $OR : RB = 3 : 2$	M1 M1 M1 M1A1 A1 [6]
	Total	**13 marks**

Part	Mark	Scheme
(a)	M1	For the correct vector statement $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB}$
	A1	$\overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$
(b)	M1	For a correct vector statement e.g $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$
	A1	For the correct simplified vector $\overrightarrow{OP} = 2\mathbf{a} + (2\mathbf{b} - \mathbf{a}) = \mathbf{a} + 2\mathbf{b}$
(c)	M1	For a correct vector statement e.g., $\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ or $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{PQ}$
	B1	For the correct vector statement for $\overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}\left(\overrightarrow{OP}\right)$ or $\overrightarrow{PQ} = -\frac{1}{4}\left(\overrightarrow{OP}\right)$
	A1	For the correct simplified vector

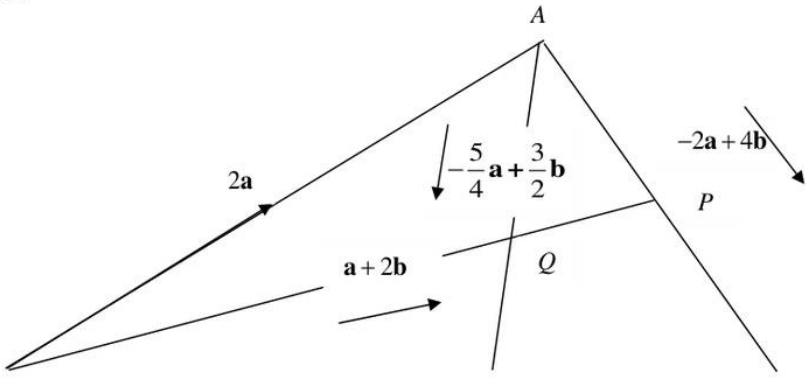
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		$\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$ or $\overrightarrow{AQ} = \frac{1}{2}(-2\mathbf{a}+4\mathbf{b}) - \frac{1}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$
(d)	In this part there is more than one route to finding the required ratio, but any route must involve $\overrightarrow{OR}$ or $\overrightarrow{RB}$ As the general principle: The first M mark is for one vector using a parameter. The second M mark is for a second vector using a different parameter. The third M mark is for equating coefficients and forming two equations in both parameters. The fourth M mark is for solving their equations. Check their working and do not allow an erroneous method. The following is using one path. Please trace their path on the sketch to check that it is valid. The path they use must involve the vector $\overrightarrow{OR}$ or $\overrightarrow{RB}$ eg., $\overrightarrow{AR} = \overrightarrow{AO} + \overrightarrow{OR}$ or $\overrightarrow{AR} = \overrightarrow{AB} + \overrightarrow{BR}$	
	M1	For the correct statement $\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$
	M1	For the correct statement for $\overrightarrow{OR}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$
	M1	For equating the two statements for $\overrightarrow{OR}$ and equating coefficients $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$
	M1	For solving their two simultaneous equations **correctly** to find the value of λ (the parameter for OR) $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \frac{8}{5}$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \ldots$
	A1	For the correct value of λ $\lambda = \frac{12}{20} = \left(\frac{3}{5}\right)$
	A1	For the correct ratio $OR : RB = 3 : 2$

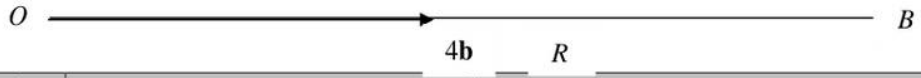
Useful Sketch



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Question	Scheme	Mark

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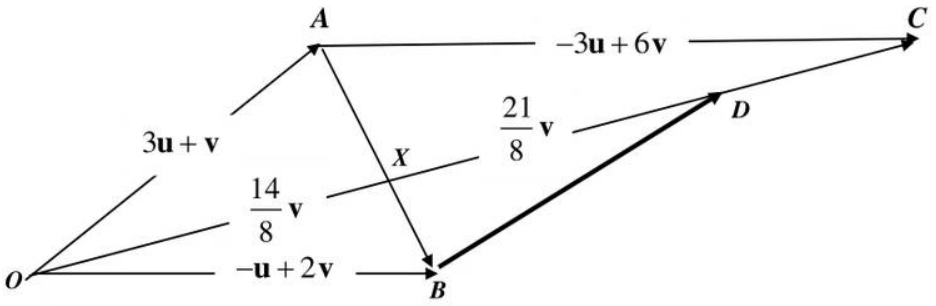
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8 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$	M1,A1 (2)
(b)	$\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$	M1
	$\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$	M1
	$\mu = 7 \qquad [\lambda = 3]$	A1 (4)
(c)	$\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$ or $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$	M1
	$\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$	A1
	$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$	M1
	$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$	M1
	$\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$	A1 (5)
		Total 11 marks

USEFUL SKETCH

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



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Part	Mark	Additional Guidance
(a)	M1	Use of $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$
	A1	$-4\mathbf{u} + \mathbf{v}$
(b)	M1	For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$
	M1	For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$
	M1	For equating for **u** and for **v**
	A1	For $\mu = 7$
(c)		**General principles for marking part (c)** For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark • Candidates will need to find vector $\overrightarrow{OX}$ [M1A1] • Candidates will need to use the ratio of bases to find $\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ **leading** to establish $\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ [M1] • Check any path they choose on the diagram below. For example, $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OX} + \overrightarrow{XD}$ or [M1] $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AX} + \overrightarrow{XD}$ If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this. • For the correct vector
	M1	For an expression for $\overrightarrow{OX}$ including an unknown constant
	A1	For $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$
	M1	For recognising that triangle OBX and triangle OBD share a base OB and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$ OR For recognising that triangle OBX and triangle BXD share a base BX and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$
	M1	For method to find $\overrightarrow{BD}$ Award this for any valid path given
	A1	$\overrightarrow{BD} = \mathbf{u} + \frac{19}{8}\mathbf{v}$

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Question number	Scheme	Marks